Project Overview & Objectives

This portfolio project addresses a critical business challenge faced by a leading retail enterprise: understanding multi-channel consumer shopping behavior to optimize marketing efficiency, maximize customer lifetime value (CLV), and improve brand loyalty.

Using historical transactional and demographic data, I built an end-to-end data pipeline to analyze purchasing patterns across various product categories, sales channels, and consumer demographics. The core objective was to answer a fundamental business question: *How can the company leverage consumer data to identify trends, improve engagement, and optimize product strategies?*

Technical Workflow & Implementation

1. Data Engineering & Wrangling (Python)

- Developed data ingestion scripts to process raw customer behavioral datasets.
- Utilized **Pandas** and **NumPy** to perform exploratory data analysis (EDA), handle structural anomalies, clean missing values, and transform the dataset for database storage.

2. Analytical Database Querying (SQL)

- Designed a structured relational schema to organize the clean retail data.
- Formulated advanced SQL queries utilizing **CTEs** and **Window Functions** to extract complex business insights, segment customers, and isolate key drivers of repeat purchases.

3. Interactive Business Intelligence (Power BI)

- Built a dynamic, stakeholder-ready executive dashboard to track critical business KPIs and purchasing trends across online and offline channels.
- Implemented custom **DAX** measures to create flexible, time-intelligent metrics for interactive data discovery.

4. Business Strategy & Stakeholder Presentation

- Translated complex technical metrics into an executive-level summary and actionable business recommendations.
- Prepared a data presentation designed to bridge the gap between technical data and non-technical business leaders to drive product and marketing decisions.